

The Ties that Bind: How Boundary Spanners Create Value in Science-Based SMEs

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How do boundary spanners, through processes of temporal sensemaking, use established and newly forming relationships to create value in science-based SMEs? In addressing this question, we examine the creation of social capital arising from the activities that boundary spanners use in establishing ‘ties that bind’ people together. We show how effective boundary-spanning activities are able to promote collaborations across scientific and related disciplines, supporting knowledge generation and innovation. Sensemaking is central to this process. We illustrate how retrospective sensemaking draws on knowledge, skills and experiences, often triggered through historical connections and personal contacts, whilst prospective sensemaking often arose in the search for new customers and in the ongoing assessment of future potential markets. We reveal how combining individual and collective boundary spanning with a continual movement between the backward glance (retrospective) and more future-oriented (prospective) sensemaking was particularly effective in generating new understandings and opportunities. We use our findings to present an inductive model of value creation that frames the dynamic interplay between individual and collective boundary-spanning activities and temporal sensemaking in the generation of social capital that acts as a knowledge resource for identifying and evaluating ideas for innovation and future pathways to value creation.

Introduction

We adopt a knowledge-based view of value creation (De Silva, Howells and Meyer, 2018) to explore how sensemaking processes build and sustain boundary-spanning relationships. We argue that these are essential to an organization’s ability to explore, acquire, retain, integrate and exploit knowledge. Sensemaking refers to how individuals ‘make sense’ of the world around them (Weick, 1995, p. 4). Our focus is on boundary spanners, who build relationships across organizational boundaries (Hettich and Kreutzer, 2021). These relationships support the development of social capital in technology and science organizations (Vlaisavljevic, Cabello-Medina and Pérez-Luño, 2016). Nahapiet and Ghoshal (1998, p. 243) define the value of social capital as ‘the sum of actual and potential resources within, available through, and derived from the network of relationships possessed by an individual or social unit’. However, ‘as a value-creation mechanism, the impact of social capital has not been fully explored’ (Edelman *et al.*, 2004, p. S67). There is also a call for further research into the complex interplay between individual boundary span-

ners and emerging collective boundary-spanning practices (Banerjee and Corredoira, 2013), and the need to better understand how individuals exchange and recombine knowledge as part of a collective innovation process that creates value (Wang, Chen and Fang, 2018).

In examining boundary spanning and processes of temporal sensemaking (Cunliffe, Luhman and Boje, 2004; Dawson and Sykes, 2019; Wiebe, 2010), in which our subjective experiences of the present influence our interpretations of the past and our expectations of the future, we draw on the work of Weick (1995). Weick, Sutcliffe and Obstfeld (2005) advocate the need for research on temporal sensemaking that facilitates shared understanding across organizational and disciplinary boundaries. The importance of collective sensemaking in this context has received relatively little attention and few studies have attempted to address ‘how individuals who hold different pieces of information are able to collectively construct new meaning’ (Maitlis and Christianson, 2014, p. 102), not only within, but also across organizations.

In addressing these concerns, we extend our research beyond the conventional focus on individual boundary

Table 1. The science-based context

Dimension	Description
Structures and institutions	The science-based context requires different organizational and institutional arrangements that support the convergence of 'science' and 'business' and facilitate 'technical advance and the diffusion of innovation' (Pisano, 2006, p. 5).
Integrating diverse disciplines	In a science-based context, innovation is generally the result of the recombination and integration knowledge and learning across diverse disciplines that require the integration of complex processes and competencies (Banerjee and Corredoira, 2013).
Risk and uncertainty	The science-based context presents 'a higher risk profile... than in other contexts' (Pisano, 2006, p. 16) in terms of commercial success.
Time horizon	The science-based context requires extended times scales to resolve fundamental uncertainties inherent in innovation before innovations can be brought to market activities (Gassmann and Reepmeyer, 2005).
Resource demands	The financial cost of science-based innovation is higher than that of other contexts, with R&D being the major component of total cost in science-based industries (Pisano, 2010).
Networks of relationships	The science-based context requires the ability to build strong networks of relationships and social capital that supports value creation (Vlaisavljevic, Cabello-Medina and Pérez-Luño, 2016).

spanners (Balogun *et al.*, 2005) and retrospective (backward glancing) sensemaking (Weick, 1995). In doing so, we examine collective boundary-spanning activities (Banerjee and Corredoira, 2013) and processes of prospective (future-oriented) sensemaking (Konlechner *et al.*, 2019) in science-based small and medium-sized enterprises (SMEs) seeking to mobilize scientific knowledge for value creation. Our findings benefit managers and leaders of science-based firms who want to better understand how boundary spanners use sensemaking to build social capital that creates value. We illustrate how the effective deployment of social capital challenges organizations to establish and develop meaningful relationships among people – 'ties that bind' – from disparate scientific and related disciplines. The specific research question we address is: How do boundary spanners, through processes of temporal sensemaking, use established and newly forming relationships to create value in science-based SMEs?

Our paper opens with a discussion on the unique context of science-based SMEs and the importance of knowledge exchange. We then review our key concepts of social capital and value creation, and boundary spanning and processes of sensemaking. We explain our methodological approach in collecting and analysing data. After presenting and discussing our findings, we develop an inductive model for value creation. The conclusion draws out our contributions, evaluating practical implications and identifying avenues for further research.

Theoretical and contextual background

Science-based SMEs

Science-based SMEs are distinct from other types of small (and larger) organizations, across a number of dimensions (Table 1). The science-based context presents 'unique challenges that require different kinds of orga-

nizational and institutional arrangements and different approaches to management' (Pisano, 2006, p. 4). There is a need to integrate diverse disciplines and pool collective knowledge to create mutual benefits and value across organizational domains (O'Cass and Ngo, 2011).

Uncertainty around the commercial benefits of innovation activities that consume large amounts of resources over prolonged periods highlights how the science-based context is often inherently more risky than other types of business enterprise (Pisano, 2010, p. 467). Pisano (2010, p. 473) highlights three concerns. First, the risk management problem – the need to encourage and reward profound risk-taking over extended time periods (Gassmann and Reepmeyer, 2005). Second, the integration problem – the need to manage knowledge integration across scientific and related disciplines. Third, the learning problem – the need to manage and encourage activities that support cumulative learning and knowledge exchange. Addressing these challenges requires understanding how retrospective and prospective sensemaking, occurring in individual and collective boundary-spanning activities, plays a crucial role in managing knowledge that creates value (O'Cass and Ngo, 2011). As Pisano (2010, p. 2) points out, science-based business 'attempts not only to use existing science but also to advance scientific knowledge and capture the value of the knowledge it creates'. He offers a broad definition of science-based business encompassing a range of organizations often not considered traditional science-based businesses, such as consumer electronics, finance and media, where technology and science converge (Sato, 2008).

We take on this broader definition that considers the importance of sharing experiences and knowledge across boundaries, arguing that this is essential to the innovation processes in science-based SMEs. These organizations are particularly reliant on the strength of

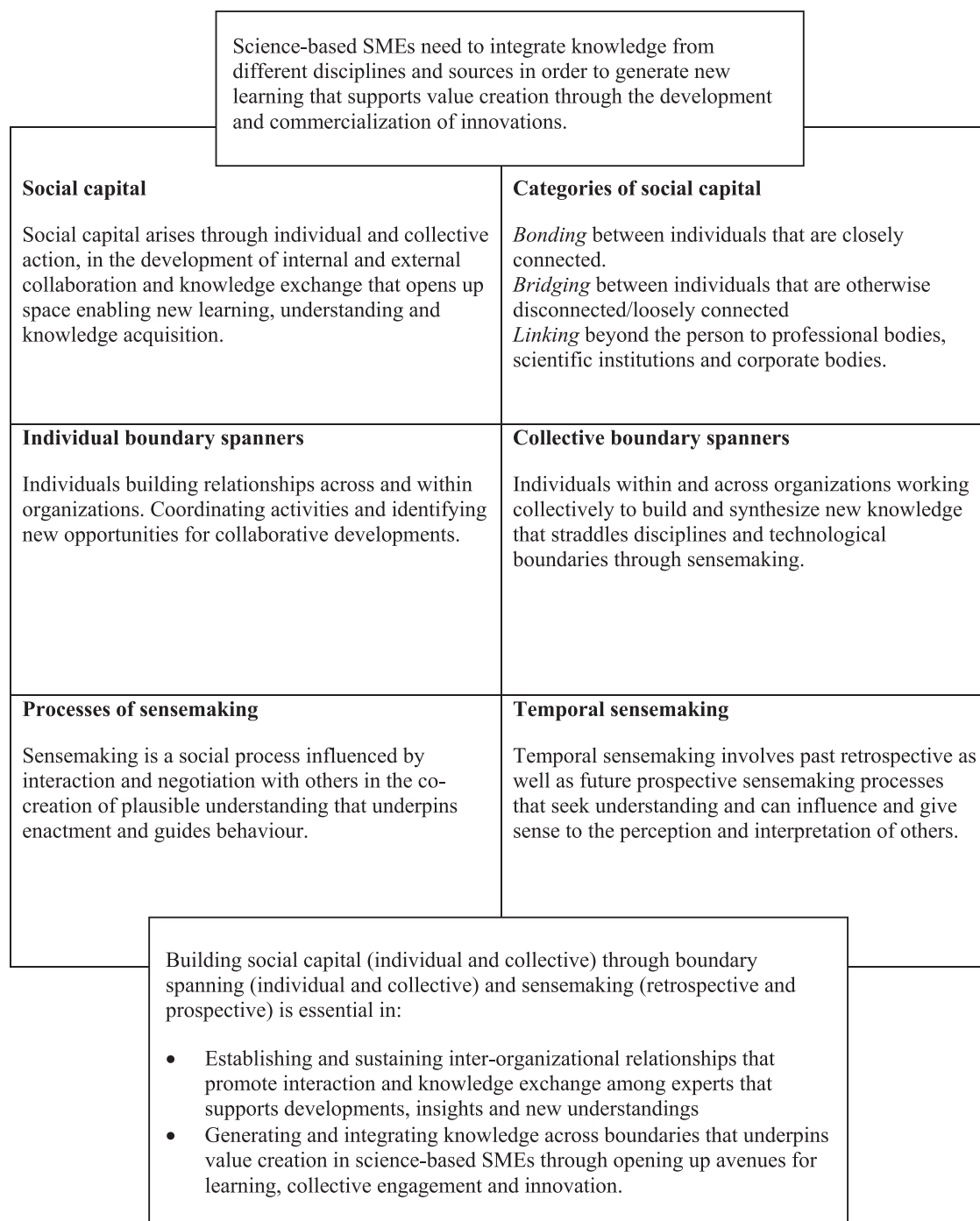


Figure 1. Components of value creation in science-based SMEs

their networks and the personal relationships that entrepreneurs form with universities, research laboratories and other science-based firms (Lee, Lee and Pennings, 2001). These relationships establish ‘ties that bind’, building social capital through collective boundary spanning and sensemaking. As outlined in Figure 1, three components of value creation – comprising social capital, boundary spanning and sensemaking – are the focus of our investigation and theoretical contribution.

Social capital and value creation

Contemporary views of science-based innovation adopt a relational view of social capital (Vlaisavljevic, Cabello-Medina and Pérez-Luño, 2016), in which the capacity to build relationships is essential for creating value (O’Cass and Ngo, 2011). Supporters of social capital theory argue that it creates value through sharing knowledge and expertise and is dependent on close

social ties (Partanen *et al.*, 2008). However, critics claim that although social capital builds strong ‘ties that bind’, it can be exclusionary (Adler and Kwon, 2002), presenting a barrier to external knowledge and ideas (Edelman *et al.*, 2004). However, most agree that value is created through innovative products, processes and services (Bowman and Ambrosini, 2000). Value can be formal (patents, trademarks, copyrights, design rights) and informal (secrecy, lead-time advantages, complexity, complementary assets) mechanisms (Desyllas *et al.*, 2018). The question arises as to whether and how science-based SMEs can leverage social capital (Pattinson and Preece, 2014) through developing intra- and inter-organizational relationships (Hettich and Kreutzer, 2021) that support knowledge exchange.

There are three broad categories of social capital. First, *bonding* social capital emphasizes existing relationships that ‘bring together people who are like one another in important aspects’ (Putnam, 2002, p. 11). Bonding social capital is about relationships amongst individuals or groups who are closely connected in some way, such as work colleagues (Newell, Tansley and Huang, 2008). Second, *bridging* social capital is about creating ties between otherwise disconnected, or loosely connected, groups and individuals (Tan, Zhang and Wang, 2015), for example with potential customers, suppliers or competitors (Putnam, 2007). Third, *linking* social capital extends the bonding and bridging distinction (Daskalaki, 2010). It involves creating new connections to individuals, groups or other corporate and institutional intermediaries, such as public institutions, universities, legal institutions and religious/political groups (Cofré-Bravo and Engler, 2019). In our research, we explore the impact of each of these three categories of social capital on value creation.

Boundary spanning and processes of sensemaking

Making sense of complexity is essential to value creation in science-based SMEs (Pattinson and Preece, 2014). In seeking to create value (O’Cass and Ngo, 2011), boundary spanners recombine and integrate knowledge in new ways (Wang, Chen and Fang, 2018). They often use peripheral knowledge to build networks and expand understanding in a given organizational context (Lindgren, Andersson and Henfridsson, 2008). Through discourse, language and text, individuals and groups continuously interact and draw on past experiences in making sense of interconnected circumstances (Weick, 1988). Boundary spanners are central to this process in their interactions with others, in explaining new ideas and connections, and in acquiring and disseminating new knowledge.

Through interactions with individuals and groups, boundary spanners seek to develop a common narrative

in the co-creation of meaningful understanding (Pattinson, Nicholson and Lindgreen, 2018). They are active players in a dynamic social process (Hettich and Kreutzer, 2021). Boundary spanners often mediate between groups with different values, beliefs, interests and power (Mehra and Schenkel, 2008), building relationships *across* and *within* organizations (Newell, Tansley and Huang, 2008). A central objective is to make and give sense to new knowledge and arrive at a common understanding. Understanding is often centred on the co-creation of a plausible narrative about what is going on around us (Boje, 2008), without necessarily seeking an ultimate truth about how the world really is (Weick, 1995). Social context is critical to sensemaking processes (Weick, Sutcliffe and Obstfeld, 2005) and boundary spanners play a key role operating at the interstices of organizations, integrating and recombining knowledge (Rosenkopf and Nerkar, 2001). They are pivotal to value creation (O’Cass and Ngo, 2011), in seeking to make sense of new knowledge and new opportunities.

At the collective level, less attention is paid to how boundary-spanning practices are combined within collaborative arrangements (Rossi *et al.*, 2022), and how boundary spanners bring together different knowledge domains to engage in collective activities that generate knowledge (Hawkins and Rezazade, 2012). Collective boundary spanning is important for science-based SMEs where knowledge exchange and recombination can create new inventions (Banerjee and Corredoira, 2013). At an organizational level, developing collective ‘combinative capability’ (Kogut and Zander, 1992) is necessary to acquire and integrate knowledge across scientific and technological boundaries (Rosenkopf and Nerkar, 2001). Central to this knowledge acquisition and integration are the processes by which boundary spanners make sense of their experiences and develop collective understanding that builds social capital (Noble and Jones, 2006), underpinning value creation (O’Cass and Ngo, 2011).

Although the relationship between sensemaking and value creation remains unclear and under-researched (Ellway and Dean, 2016), it is central to our study on how boundary spanners create value in science-based SMEs. Our focus is not simply on sensemaking as a retrospective process, the Weickian ‘backward glance’ (Weick, 1995), where sensemaking processes are triggered retrospectively. We also explore sensemaking as a prospective process (Konlechner *et al.*, 2019), where people reinterpret the past in looking forward to future possibilities (Dawson and Sykes, 2019). This draws attention to the importance of making sense of future potential avenues of innovation, where ‘The attention and concern of people is primarily directed at events that may occur in the future’ (Rosness *et al.*, 2016, p. 55). In addition to countering the lack of attention to

prospective sensemaking (Konlechner *et al.*, 2019), our research also sets out to accommodate a more dynamic temporal perspective in the interplay and interweaving of these continuous processes (Tsoukas *et al.*, 2020).

We aim to show how, in order to make sense of the world around them and inform action, people draw on the past, present and future through processes of temporal sensemaking (Dawson and Sykes, 2018). Our central aim is to uncover how boundary spanners, through these processes of sensemaking, create value in science-based SMEs (Figure 1 presents an overview of the core components we identify as important in our investigation).

Methodology

Research design

This study draws on data from qualitative fieldwork, allowing an in-depth exploration of the phenomenon of interest. Semi-structured interviews provided the flexibility to gather rich data essential to answering our research question. Data analysis utilizes a grounded theory approach using first and second-order coding to present our findings (Strauss and Corbin, 1990). These show how boundary spanners, through sensemaking processes and interactions with others (individually and collectively) (Fan and Dawson, 2022), promote the development of social capital.

Case selection and description

We used theoretical sampling to select the most theoretically relevant case organizations. With the help of one of the United Kingdom's National Science Parks, we generated an initial list of potential science-based SMEs. We then conducted online searches, exploring each organization's website to assess its suitability with our definition and research aims. We applied two criteria to assess suitability: (1) the organization fits within Pisano's definition of a science-based business that captures value from scientific knowledge; and (2) organizational size and scope fits within the European Union definition of a small and medium-sized enterprise. This means each organization has fewer than 250 employees, a turnover of less than 50 million euros and an annual balance sheet not exceeding 43 million euros. Table 2 details each organization's key characteristics. The organizations represent a range of science-based disciplines encompassing chemical processes, biotechnology, photocatalysis, nanotechnology, radiation detection, gas analysing and marine fuel analysis. The diversity in terms of size and business activities helped elicit a wide range of responses, enabling us to capture the stories behind participants' experiences.

Data collection and sample

We conducted 23 in-depth interviews with a range of individuals in the seven companies (Table 3). Interviewees were employed in a range of scientific, engineering and commercial roles to ensure we captured a diversity of perspectives. The semi-structured interview protocol included 10 questions. Each interviewee was provided with a copy of the questions in advance. The initial questions provided a starting point for each interview, but enabled a flexible approach, allowing the interviewer to deviate from the 'script' and pursue other interesting responses. Interviews were conducted by the lead author and varied in length from 30 to 65 minutes. Interviews were digitally recorded and subsequently transcribed. The interviewer also made handwritten field notes during observations of work practices. The observations consisted of site visits where the interviewer was shown round the facilities. Each visit lasted between 1 and 2 hours, depending on the time available. In addition to the interviews, field notes and site visits, the analysis was supported with archival materials used to gather contextual background information to help us understand the histories of each case organization. Given the narrow aim of our study (how boundary spanners create value in science-based SMEs), the dense sample specificity (participants with a broad knowledge base and experience of the research context) and quality of the dialogue (in-depth, rich data), we determined the saturation point to be when the data from additional interviews produced no new information (Saunders and Townsend, 2016).

Data analysis

In Figure 2, we present the coding structure where initial first-order codes were grouped into second-order micro-foundations and then into the final aggregate dimensions. We employed an inductive, grounded approach to developing a coding structure that supports our social constructionist position and helps us to understand the meanings and nuances associated with the sense-making processes of boundary spanners. We argue that knowledge is socially constructed from the social and physical environment and the histories of the people involved. A grounded approach possesses a discernible thread of social constructionism (Strauss and Corbin, 1990) that allowed us to focus our analysis on the means by which interviewees make sense of their experiences (Gioia, Corley and Hamilton, 2013). In following the Gioia methodology (Gioia, Corley and Hamilton, 2013), we developed a comprehensive compendium of first-order (informant-centric) codes, which we then organized into our second-order (theory-centric) codes. These were then distilled into our final overarching theoretical dimensions. Gioia advocates reporting both first-order and second-order analyses that serve as the basis

Table 2. Participant organizations

Company	Established	Employees	Turnover	Sector	Commercial activities	Geographical scope
A	1992	80	£5 m	Chemical processing	Company A offers bespoke outsourcing solutions to the speciality chemicals industry. Its products include flame retardants, agrochemicals, biocides, herbicides and animal feed additives. It has specialist knowledge in a variety of chemical processing methods such as granulation, extrusion, roller compacting, impregnation and coating. It also specializes in the size reduction of solids by crushing, milling and micronizing as well as offering support facilities, packing services, quality systems and expert technical management back-up.	National
B	2002	14	£1.2 m	Biotechnology	Company B manufactures a range of in-vitro diagnostic products for use in the detection, prevention and monitoring of medical conditions related to haemostasis and platelet function. Used in combination with an electronic instrument, the company's products can determine how quickly or slowly blood clots for Warfarin patients. Warfarin has a wide range of applications, including treating patients who have had a heart valve replacement, deep vein thrombosis or atrial fibrillation (irregular heartbeat).	International
C	2009	2	Not available	Wastewater treatment	Company C specializes in water treatment systems that use a process called 'photocatalysis' to destroy a broad spectrum of organic contaminants in water. Unlike traditional water treatment systems such as filters, granular activated carbon or membranes, which work by concentrating contaminants for later disposal, photocatalysis systems efficiently destroy organics in situ, with little or no extra disposal costs, leaving clean, drinkable water.	International
D	2008	2	Not available	Nanotechnology	Company D produces customized nanoparticulate dispersions and ultra-stable emulsions for use in a wide range of industries including paints, food processing and medicine. The nanotechnology developed by the company allows hard particles to be processed to produce stable dispersions of nanoscale particles at a rate that is industrially significant. The technology is particularly effective at creating emulsions with nanoscale droplets at a very wide range of viscosities.	National

Table 2. (Continued)

Company	Established	Employees	Turnover	Sector	Commercial activities	Geographical scope
E	2003	60	£2.4 m	Radiation detection equipment	Company E manufactures semiconductor materials within the cadmium telluride family, which have a variety of applications in the areas of medical imaging, security screening, industrial inspection and space exploration. Although originally set up to commercialize the production of zinc cadmium telluride crystals, the company now produces a range of products to deal with liquid-based terrorism threats in aviation security and border control.	International
F	1981	40	£6 m	Gas sensor and analyser equipment	Company F designs and manufactures gas sensors and analysers. The company offers a diverse range of products and services including for sports and commercial diving, life support for divers in diving bells and chambers working offshore, industrial and laboratory safety (e.g. industrial safety equipment for toxic, flammable and inert gases) and hospitality, pubs and breweries, where it manufactures carbon dioxide sensors. It also provides life-support solutions for naval diving and submarine atmosphere monitoring.	International
G	2003	46	£4 m	Marine fuel analysis and testing	Company G is a marine fuel testing company and provides on-site and laboratory-based analysis and testing on the quality of marine fuels used by all types of seagoing vessels. In addition to these field and laboratory-based services, the company also offers full investigative analysis to examine any damage caused to ships' operation by fuel contaminants which are non-naturally occurring components of the residual/distillate bunker fuel.	International

Table 3. Interviewees

Company	Interviewees	Length of interview	Reason for inclusion in study	Additional archival material
A	Operations Director	50 minutes	Science-based SME in the chemical processing sector	Company website
	Purchasing Manager	35 minutes		Product brochures
	Commercial Director	50 minutes		Technical product specifications
	Finance Director	45 minutes		Marketing and promotional materials
B	Managing Director	65 minutes	Science-based SME in the biotechnology sector	Press releases
	Technical Manager	35 minutes		Company website
	R&D Manager	45 minutes		Sister company website
	R&D Scientist	50 minutes		Online product catalogue
C	Managing Director	40 minutes	Science-based SME in the water treatment (photocatalysis) industry	Quality assurance certificates
	Technical Director	45 minutes		Press releases
D	Managing Director	50 minutes	Science-based SME in the nanotechnology sector	Company website
	Technical Director	37 minutes		[W] Innovation Centre website
E	Materials Manager	45 minutes	Science-based SME in the semiconductor industry	[N] Central website
	New Technology Manager	60 minutes		Press releases
	Chief Technical Officer	60 minutes		Company website
	Technical Manager	50 minutes		CEO podcast
F	Managing Director	40 minutes	Science-based SME in the gas sensors and analysers sector	Company YouTube channel
	Design Engineer	30 minutes		Customer testimonials
	R&D Manager	35 minutes		Press releases
	Service Manager	40 minutes		Company website and blog
G	Technical Director	40 minutes	Science-based SME in the marine fuel testing industry	Product brochures
	Operations Director	30 minutes		Photographs
	Customer Services Manager	40 minutes		Press releases

for both our static data structure (Figure 2) and our inductive model (see Figure 4 later), which fits with our overall methodological approach.

First, to identify the initial first-order codes (Strauss and Corbin, 1990), the lead author independently reviewed the interview transcripts to identify initial codes in the raw data and to categorize them according to Van Maanen's (1979) coding process. Using open coding, the data were sorted into data incidents to identify initial differences and similarities (Strauss and Corbin, 1990) connected to boundary spanning, sensemaking and social capital. Before moving from initial descriptive codes to more inclusive second-order codes, the second author then challenged the first-order codes and through an iterative process and detailed discussion, the authors achieved consensus about the final data

incidents to be included. This iterative process identified the initial 34 first-order codes. Second, in developing the second-order codes, the authors searched for connections across the case organizations between and amongst the first-order codes, grouping them into six second-order concepts. At this stage, we were interested in how sensemaking processes were deployed to build relationships and how social capital was developed through boundary spanning. We continually reviewed the data to ensure reliability and validity (see Table 4), enabling us to confirm that our understanding was trustworthy and credible (Gioia, Corley and Hamilton, 2013).

Third, we distilled our second-order codes further, grouping them into the final three aggregate dimensions of value creation. We developed a cross-case

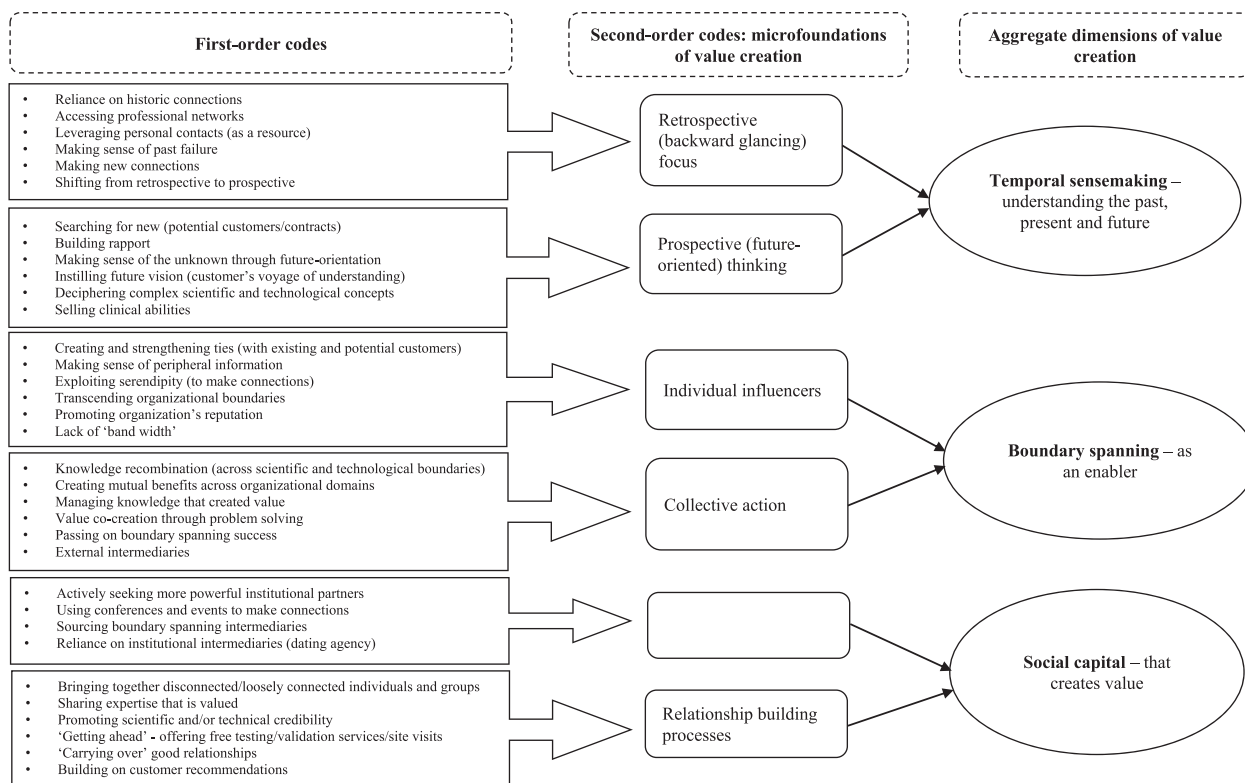


Figure 2. Data structure

analysis to compare and contrast insights emerging through our data analysis and coding process to identify consistent patterns and themes (Eisenhardt, 1989) that relate to how boundary spanners use sensemaking to engage others in collective understanding that helps to build social capital, underscoring value creation in science-based SMEs. Using cross-case analysis enabled us to mobilize data incidents across the case organizations. Thus, we were able to develop a clearer understanding of the similarities and differences in how the sensemaking processes of boundary spanners build social capital that creates value. In line with our inductive approach, we have tried 'to give the most germane categories memorable labels or descriptors (while trying hard to retain informant terms, if at all possible, even at the theoretical level)' (Gioia, 2021, p. 25). To this extent, our analytical approach provides an acceptable balance between categorization and inductive analysis. This process enabled us to develop a deep understanding of the common patterns and idiosyncratic characteristics across the case organizations.

Value creation in drawing on social capital arising from sensemaking processes and boundary-spanning activities

Using our data structure (Figure 2), we develop a narrative-theorizing approach. We present quotations

from respondents that relate to the identified microfoundations and aggregate dimensions of value creation. These quotations are used as exemplars in our analysis and cross-referenced and presented in Tables 5–7.

Temporal sensemaking – understanding the past, present and future

Our empirical data draws attention to the importance of temporal sensemaking. Our analysis shows how, through understanding past, present and future, boundary spanners leverage future opportunities and innovations that create value for the organization. Within and across organizations, boundary spanners establish and develop relationships in which sensemaking processes act as a 'springboard into action' (Weick, Sutcliffe and Obstfeld, 2005, p. 409).

Retrospective (backward-glancing) focus. For some individuals, the focus of their sensemaking was on the past (a retrospective backward glance), to make plausible sense of past events and experiences. For example, one manager described his historic network in terms of keeping in touch with 'old friends', talking about new technologies. This process enabled him to make sense of how new science and technology was being applied in other sectors (Table 5, quote 1). Reliance on historic connections (Pattinson, Nicholson and Lindgreen,

Table 4. Reliability and validity criteria (adapted from Lincoln and Guba, 1985)

Quality criterion	What we did to ensure reliability and validity
Credibility (similar to internal validity)	To guarantee credibility of the analysis of the interview data, we triangulated our evidence against the archival data sources from the case organizations (see Table 1). By cross-referencing multiple sources, we were able to ensure the trustworthiness of our evidence.
Dependability (similar to reliability)	To ensure dependability, we asked a senior academic colleague, not connected to the research, to audit the data, to confirm the accuracy of the analysis and to ensure that the data supported the findings. This provided valuable insights that helped us to better articulate our findings.
Confirmability (similar to objectivity)	To confirm our confidence in the quality of the data analysis, we adopted a reflexive approach. We used immersion and crystallization techniques to continually review the data and emerging themes, enabling us to build a credible understanding of the data (see Borkan, 1999). Immersion involved reading, re-reading or examining individual portions of the data in detail. Crystallization involved temporarily suspending data analysis to reflect on the themes identified in the immersion process.
Transferability (similar to external validity)	Although we do not make any claims regarding the generalizability of our evidence, we have provided detailed descriptions of the case organizations and interviewees that could support other researchers studying boundary spanning across different contexts.

2018) accessed through professional networks was further evidenced in Company E when seeking new knowledge (Table 5, quote 2). Similarly, Company G relied on personal connections to test its innovative abatement technology for cleaning marine fuel (Table 5, quote 3). In one example, the personal networks of ex-colleagues were accessed to revisit a previously abandoned business opportunity. Through interacting with historic connections, the Technical Director of Company D adopted a ‘backward glance’ to help make sense of past failure to establish a new science-based venture (Table 5, quote 4). In contrast, Company C, an early start-up based in an incubator unit at the [W] Innovation Centre, did not have the historic industry connections to support the wider interactions and collective sensemaking necessary for building social capital that creates value. However, Company C was able to build linking social capital and make new connections through an alternative pathway with the support of the Centre Manager (Table 5, quote 5). He was able to create value for Company C by bringing companies together in an innovation forum which served a similar purpose.

What is interesting about these illustrative examples is how newly established relationships develop into a network of support for the collective sensemaking of new technologies and techniques. Networks from the past were also drawn upon to make sense of developments in known science-based areas, as well as areas of new scientific knowledge. These processes are interactive, reciprocal and ongoing, in which participants influence their collective sensemaking in forming new interpretations that create value by supporting learning and knowledge acquisition. These networks also provide an important resource when attending to future challenges and avenues of opportunity. For example, the Technical Director of Company C, through processes of collective sensemaking within the [W] Innovation Centre’s innovation forum, was able to begin to shift from a more

reflective and retrospective sense of what was towards a more future-oriented (prospective) view. Using temporal sensemaking helped him recognize new potential areas for value creation (Table 5, quote 6).

Prospective (future-oriented) thinking. The findings revealed individuals adopting future-oriented thinking, using prospective sensemaking to synthesize scientific knowledge to shape future outcomes (Konlechner *et al.*, 2019). For Company B, this meant using its expertise to provide support and advice to help customers make sense of how they might use its products. The company hoped that building rapport would help shape positive future outcomes (Table 5, quote 7). These future-oriented boundary spanners also act as change agents, seeding new ideas and promoting new potential pathways. For example, Company E sought to build rapport by promoting its own unique expertise (Table 5, quote 8). In this case, value creation was through the successful promotion of the company’s unique base of scientific knowledge. This was partly predicated on the CEO’s ability to decipher complex scientific and technological concepts and explain these in a non-technical way to potential customers. Adopting a future-oriented approach enabled him to recognize that customers (who often did not have a deep understanding of scientific knowledge) need help in making sense of the unknown (Table 5, quote 9). The Technical Manager of Company E illustrates elements of this when he explains his role as a guide, taking his customer on a voyage of understanding to find a solution (Table 5, quote 10). In adopting a future-oriented view, the Technical Manager uses prospective sensemaking to create value by building a shared vision to a pathway for customers to achieve these outcomes.

Similarly, the Chief Technical Officer of Company E recognizes the value of reinforcing the company’s scientific and technical capabilities by helping decipher complex scientific knowledge for customers (Table 5,

Table 5. Illustrative quotes: temporal sensemaking – understanding past, present and future

Quote No.	Exemplar quotes	Interviewee	First-order code
Retrospective (backward-glancing) focus			
1	[I've] got a lot of old friends in different business so I spend a lot of time talking to people about technologies in other areas.	Chief Technical Officer, Company E	Reliance on historic connections
2	Old networks invariably. I would say old networks from lots of different places... but certainly it's somebody knew somebody, knew somebody, you know, and it tends only to be two degrees of separation. If you look at these people, they would be on my LinkedIn for example.	Materials Manager, Company E	Accessing professional networks
3	We were fortunate, at the start, to get a very large shipping company who knew us well from our previous [jobs] and because they knew us personally... they were willing to try out our ideas and give us feedback to improve our product.	Technical Director, Company G	Leveraging personal contacts (as a resource)
4	I kept in touch with some colleagues of mine at university and this opportunity or idea came up. The original concept didn't work out, but the idea was still there, strong enough that I managed to get together with my other co-founder, an ex-colleague, and [from that discussion] we developed the company.	Technical Director, Company D	Making sense of past failure
5	We had our first sort of get-together of water treatments... water monitoring company conference rather, where [the Centre Manager] contacted a few companies in the area, mostly companies that didn't directly compete with each other. And we all gave a little presentation on our own companies and had a network[ing] event after that. And we have some meetings with some of those companies next week, actually.	Technical Director, Company C	Making new connections
6	I thought perhaps [support] through the [W] Innovation Centre would be adequate, but that wasn't enough. And I think once I made contact with the engineering people on the site... it took longer to get in touch with the right people than I'd anticipated... I had to look to my old contacts to help in developing the commercial product.	Technical Director, Company C	Shifting from retrospective to prospective thinking
Prospective (future-oriented) thinking			
7	We're looking to make our [products] easier for nurses to use. It's been used by health care professionals in the past. Now nurses need to understand it so we're taking steps to make it easier for them.	Managing Director, Company B	Searching for new (potential customers/contracts)
8	I think [the CEO] helps a huge amount... [promoting the company], he has far more involvement... speaking to the right people and being responsive. It's the rapport thing, isn't it?	Materials Manager, Company E	Building rapport
9	A lot of the stuff we do, no-one knows how to do. Very little of what we do is 'building the house', where you put the bricks in, you build the floors, you put all the wiring in. Very little of that is what we do. It's a lot of unknowns, so you have to [explain this] at most steps along the way.	New Technology Manager, Company E	Making sense of the unknown through future orientation
10	I see that my job is to take [the customer] on this voyage and explain why my vision is the right one.	Technical Manager, Company E	Instilling future vision (customer's voyage of understanding)

Table 5. (Continued)

Quote No.	Exemplar quotes	Interviewee	First-order code
11	You've got to be under the skin of what your customer's doing in every area. You've got to understand their application really... I mean it's a well-worn phrase but really what you have to do is provide solutions for people. It's not really... the product you're selling, you're selling a solution to their problem. So, you know, you can't sell a solution to their problem if you can't understand what their problem is.	Chief Technical Officer, Company E	Deciphering complex scientific and technological concepts
12	I didn't set out to be a contracting manufacturer. We really did set out to try and sell a range of products that would sell as [Company B], for us to work through a range of distributors, and we would have been very happy with that. When people started to knock on our door asking to use our scientific expertise and having seen how aggressively that has grown over the last few years, then that was certainly a business-changing event for sure.	Managing Director, Company B	Selling clinical abilities

quote 11). Company B, in addition to being an original equipment manufacturer (OEM) of its own products, sought to license its innovations to other manufacturers (Table 5, quote 12). However, it was surprised to find that it had greater commercial success through collaborations with other companies than through selling OEM products. By selling its 'clinical ability', it created better value through contract manufacturing on behalf of customers. This success relied on the Managing Director's ability to communicate effectively with prospective customers. For example, in the hope of gaining future business, the Managing Director engaged in prospective sensemaking, creating a plausible narrative about how the company can serve as an important future component to business development through its expertise and scientific capabilities.

In these examples, retrospective and prospective sensemaking was a critical enabler of boundary spanning. Although we draw on illustrations that emphasize a past or future orientation, there is a dynamic temporal interplay in which past knowledge and interpretations inform future avenues of opportunity to create value. This in turn shapes sensemaking and enactment in an ongoing present. This dynamic process does not follow a fixed, linear sequence of events. Our fieldwork data highlights a temporal interweaving, as interpretations of the past and expectations of the future feed into a continually reconstituted understanding of a moving present.

Boundary spanning and processes of sensemaking

Our findings reveal the importance of boundary spanning at the individual and collective level in influencing processes of sensemaking.

Individual influencers. Individual boundary spanners draw on their scientific knowledge and expertise in providing advice to customers that develop ties and influence sensemaking. For example, the Technical Manager of Company B explains how the Managing Director uses his position as an expert in his field as an opportunity to gain new customers through helping them understand how to use competitors' products (Table 6, quote 13). As the Technical Manager reflects (Table 6, quote 14), providing peripheral knowledge on competitors' products enabled the Managing Director to exploit serendipitous opportunities (Garud, Gehman and Giuliani, 2018) to create value by developing bridging ties to new customers.

Similarly, when visitors from 'Company T' came to view the vacant business unit next door, the Managing Director of Company B took the opportunity to introduce himself and show them around his own unit. It soon became clear that the two companies had complementary areas of scientific knowledge and expertise, with potential to create value through collaboration (Table 6, quote 15). As a result of this initial meeting, the Managing Director was able to bring together individuals in 'Company T' and [C] University to solve an issue the university was having in scaling up an innovation (Table 6, quote 16). In offering advice across organizational boundaries, individual boundary spanners can develop contacts, influence others and create ties to future potential customers. The Operations Director of Company G, for example, uses his industry experience to promote and enhance the organization's reputation (Table 6, quote 17). This supports Moran and Ghoshal's (1996) view that value creation arises from the combination and exchange of resources. In this case by developing and influencing relationships through drawing on an individual's scientific knowledge and expertise.

Table 6. Illustrative quotes: boundary spanning and processes of sensemaking

Quote No.	Exemplar quotes	Interviewee	First-order code
Individual influencers			
13	[The Managing Director has] helped out a hospital. They're not a customer but they couldn't get the support from their current supplier. So because... he's known in this sort of field, the contact [at the hospital] says 'look, I'm using this reagent, although it's not yours, this is happening to my patients, or these are the results – and it is not good'. So [the Managing Director] basically takes the data, has a look at it and says 'you can do this, this, this or this'. He hasn't tried to force them to buy our reagent or anything. He's just helped them out.	Technical Manager, Company B	Creating and strengthening ties (with existing and potential customers)
14	... so [The Managing Director] basically takes the data, has a look at it and says 'you can do this, this, this or this'. He hasn't tried to force them to buy our reagent or anything. He's just helped them out.	Technical Manager, Company B	Making sense of peripheral information
15	[Company T] came here to visit... because they're thinking about moving next door. We just got on really well and we realized that there's potential areas of collaboration, because we're from the same area but different products.	Managing Director, Company B	Exploiting serendipity (to make connections)
16	[Company T] produces novel research entities for pharmaceutical research based on peptides. It's a very good business. I just didn't know it was in [this town] that's all. We now keep very much in touch. I mean, I have a link with [C] University's Department of Biochemistry and a researcher down there is struggling to scale up his peptide synthesis, so we arranged a meeting with [Company T] because they can do it.	Managing Director, Company B	Transcending organizational boundaries
17	Our reputation just spread amongst the people that we knew from [our] previous jobs... and [the Operations Director]... he went out and visited all our [previous] contacts to promote our new fuel analysis technology.	Technical Director, Company G	Promoting the organization's reputation
18	These days I tend more to work with a limited set of people. I don't have the 'band width' to get involved. I mean the organization has grown and my role has changed so I don't have the band width.	Chief Technical Officer, Company E	Lack of 'band width'
Collective action			
19	We would class... innovation as finding solutions to customers' problems. Also, looking for areas where we can introduce different technology to the areas that we are already reasonably expert in.	Operations Director, Company A	Knowledge recombination (across scientific and technological boundaries)
20	The most successful ideas... start from the individual and then they successfully persuade others to share that motivation, that idea, that concept, and then get some momentum behind it.	Managing Director, Company F	Creating mutual benefit across organizational domains
21	I think we all try to have a close partnership relationship... it's very important... we're selling product that's quite complicated We've probably got some stuff which is moving down to catalogue purchase and certainly it's our hope to make products that become a catalogue purchase, but I think in most cases we're a long way away from that. Our customers therefore rely on our expertise and advice to find suitable solutions.	Chief Technical Officer, Company E	Managing knowledge that created value

Table 6. (Continued)

Quote No.	Exemplar quotes	Interviewee	First-order code
22	Problem solving is what you're trying to do. You're trying to solve a problem for somebody... and that then becomes understanding the problem, then it becomes trying to design the solution with the customer.	Technical Manager, Company E	Value co-creation through problem solving
23	Now when people come and visit us it's not just 'you and me'. I make sure they meet everybody; more people are involved in the meetings. They start to build their own [connections].	Managing Director, Company B	Passing on boundary spanning success
24	We'd worked with [company 1] when they developed some technology in conjunction with [company 2] at the University of [L], through [N] Central. So as a team we were able to develop the algorithms and models to say, 'we need "x" amount of energy to split these particles up'. So now you can actually say 'I need "x" amount of energy', and they're perfect as a partner to go along and say, 'build me something that'll give me "x" amount of energy'.	Managing Director, Company D	External intermediaries

For staff operating at a senior strategic level, maintaining effective individual boundary-spanning activities becomes more difficult as their contact with external people diminishes. One interviewee, who began his career as a physicist before moving into management, referred to the phenomenon as lack of 'band width' (Table 6, quote 18). This finding suggests a significant reduction in the ability of senior managers to participate in effective boundary spanning that creates value as they take on more strategic roles.

Collective action. Our findings illustrate how individual boundary spanners sought collective understanding in their search for innovations. Collective understanding arose through people engaging in social interactions (e.g. Company A) that provide meaning and a sense of opportunity in pursuing research that supports knowledge recombination (Table 6, quote 19) across scientific and technological boundaries (Banerjee and Corredoira, 2013; Marshall and Rollinson, 2004). In the case of Company F (Table 6, quote 20), these sensemaking processes are also about enactment, steering collective action towards potential opportunities for creating mutual benefits (Hettich and Kreutzer, 2021). This illustrates how individuals' scientific and technological expertise serves as a bridge for creating mutual benefits across organizational domains.

Our fieldwork data also demonstrates how collective action plays a crucial role in managing knowledge that creates value for customers. This was evident in the way customers relied on Company E to help them make sense of complex products (Table 6, quote 21). The core business activity of this company pivots on its ability to create value by identifying the most appropriate technological solutions to meet its customers' needs. Through collective boundary-spanning activities,

Company E was able to develop accessible pathways to solving problems that the customer understood. This supports a process of value co-creation (Table 6, quote 22) through joint problem-solving activities with customers (Hettich and Kreutzer, 2021), as well as enabling knowledge recombination to create new technological solutions (Banerjee and Corredoira, 2013).

Our findings spotlight many examples of how collective boundary-spanning activities bring together stakeholders including employees, customers, suppliers and competitors in various forms of value co-creation. They also illustrate how knowledgeable boundary spanners often try to pass on their learnings to colleagues. For example, the Managing Director of Company B had learned the importance of effective boundary spanning and wanted to pass this on to his staff. He encouraged staff to use their own scientific knowledge and expertise to emulate his boundary-spanning success with customers (Table 6, quote 23). He encouraged the company's R&D Scientist to extend his purely scientific role in taking on board the commercial aspects of the business, becoming a more important knowledge resource in managing the company's expanding customer base.

Perhaps unsurprisingly, there was a general absence of boundary-spanning capability in smaller start-up SMEs (Companies C and D). This was largely due to their historical positioning as new market entrants with few industry contacts. For Company C, the [W] Innovation Centre compensated for its founder's lack of boundary-spanning capability. The centre acts as an intermediary, *ex post facto*, in facilitating value-creating connections with other companies. Similarly, Company D attempts to compensate for its lack of individual boundary-spanning capability by connecting with a UK government-funded nanotechnology project, [N]

Table 7. Illustrative quotes: Social capital – that creates value

Quote No.	Exemplar quotes	Interviewee	First-order code
Institutional intermediaries			
25	So, eventually... after maybe a year and a half of knocking on the US government's door, because they are one of the key people who [fund] this type of technology, we opened up a funding contract with them and that's been the backbone of our materials development technology in the last two and a half years.	Chief Technical Officer, Company E	Actively seeking more powerful institutional partners (mediating social capital)
26	[N] Central run a conference once a year just down in London, mid-to-early June... we're able to meet up with companies we'd find hard to connect with outside of the [N] Central project.	Managing Director, Company D	Using conferences and events to make connections (linking social capital)
27	We're employing this other person on a few days a month. Now he's sort of opening doors, and he might have the initial meeting and then call us to go and meet the [potential] clients as that's gone ahead.	Managing Director, Company C	Sourcing boundary-spanning intermediaries (combining bonding and linking social capital)
28	[N] Central was essentially a dating agency, finding people with problems and introducing them to people who could solve the problems.	Managing Director, Company D	Reliance on institutional intermediaries (dating agency) (linking social capital)
Relationship-building processes			
29	He understands the technology and that's really what gives him his unique levels of focus. You've got to be able to help the customer and, because... the technology's quite difficult [to understand for] most of the sales people, they can help the customer a little, but they can't really help them as much as maybe [the CEO] can.	Chief Technical Officer, Company E	Bringing together disconnected/loosely connected individuals (bridging social capital)
30	We've got the commercial expertise, we have the technical expertise [and] we have support services, like the laboratory, the compliance side and a very strong management team. I think this is something people doing innovation value.	Operations Director, Company A	Sharing expertise (bonding social capital)
31	We, if we're dealing with a new customer, we like to bring them here so that, one, they can see the facility, and [two] they can meet the people. I think it's fair to say that when people do come along they are generally very impressed with the management team that we have. Essentially, we've got a mix of engineers and chemists, which is a good combination for this sort of business. But in addition to that, we have people like our compliance manager, who handles all the health and safety, and environmental issues, and I think people see that we have some substance.	Commercial Director, Company A	Promoting scientific and/or technical credibility (bridging social capital)
32	In certain cases what we might do is offer a kind of validation that our system might work by a sort of a... my co-founder hates this word, but a 'free test'. She doesn't like the word free (complimentary) test.	Technical Director, Company C	Getting ahead – offering free testing/validation services/site visits (bridging social capital)
33	I have extremely good relationships with everyone I work with, that's why we're successful. And I'm carrying over that good relationship to everybody in [Company B].	Managing Director, Company B	Carrying over good relationships (combining bonding and bridging social capital)
34	I think the main thing is instilling confidence in our customers. I mean... if you build up that relationship and they will pass on that information and discuss the company, and other people will approach us almost on [that] recommendation.	Design Engineer, Company F	Building on customer recommendations (combining bonding and bridging social capital)

Central (a collective alliance of leading nanotechnology organizations set up to support the nanotechnology sector). As the Managing Director explained, working through an intermediary such as [N] Central enabled Company D to exploit linking social capital, connect with other nanotechnology companies and engage in collective action (Table 6, quote 24). [N] Central enabled Company D to connect with other organizations, helping it create value by mobilizing its scientific knowledge base. Individual boundary spanners and intermediaries were thereby able to transcend organizational barriers, strengthening existing ties to customers whilst also establishing ties to new potential customers.

Social capital and the creation of value

Sensemaking was integral to the boundary-spanning processes necessary to develop individual, collective and institutional relationships (Cofré-Bravo and Engler, 2019). Our data highlights how effective boundary spanners, individually and/or collectively, were able to create value from scientific knowledge, enabling them to build relationships rich in social capital.

Institutional intermediaries. Our findings show how some companies actively seek out more powerful partners. For example, the CEO of Company E had been particularly active within a forum organized by the US government, promoting the company's scientific knowledge base in the development of zinc cadmium telluride (used extensively in the manufacture of radiation-detection equipment). It was through these boundary-spanning activities that Company E was able to create value by gaining knowledge and financial support from a more powerful institutional partner, in this case the US government (Table 7, quote 25). As the Chief Technical Officer observed, the CEO used the company's scientific knowledge to mediate a relationship with the US government, creating a tie to a powerful institution that serviced the building of social capital (Partanen *et al.*, 2008).

Institutional intermediaries such as the [W] Innovation Centre and [N] Central project played a significant role in helping the smaller start-up SMEs (Companies C and D) mobilize their scientific knowledge bases to expand and exploit linking social capital. For example, [N] Central hosted an annual conference bringing together nanotechnology-focused SMEs (Table 7, quote 26). Company C was able to leverage its bonding social capital with the [W] Innovation Centre manager, to source a new part-time member of staff, an ex-CEO with extensive industry connections. The ex-CEO acted as a bridge, a boundary-spanning intermediary, creating value by making introductions and building relationships between potential customers and the founders of Company C (Table 7, quote 27). As a start-up with lim-

ited boundary-spanning capabilities, Company D was reliant on its institutional connection with [N] Central to build connections. In essence, [N] Central acted as an intermediary, a 'dating agency', making introductions to potential customers seeking new scientific knowledge to solve problems (Table 7, quote 28). Finding people with problems and introducing them to people who could potentially solve their problems was central to the creation of value in Company C (Hettich and Kreutzer, 2021; O'Cass and Ngo, 2011). As an institutional actor (Cofré-Bravo and Engler, 2019), the [N] Central project was a crucial mechanism in linking social capital. This enabled Company D to showcase its scientific knowledge in building valuable institutional relationships with potential customers. Company C was also active in using the [W] Innovation Centre as an institutional intermediary to provide support in its value-creation activities.

Relationship-building processes. Relationship building within and across organizations was essential in building social capital at individual and organizational levels. The CEO of Company E, for example, was originally an academic with a PhD in physics. However, fellow board members also saw him as a credible business person, able to combine extensive scientific knowledge with well-developed business acumen. As the Chief Technical Officer explained, the CEO of Company E was able to draw upon his deep understanding of the company's proprietary technology and explain it in a way that sales staff could more easily convey to customers (Table 7, quote 29). The CEO acted as a boundary spanner, creating value through building relationships for bridging social capital between the company's sales staff and its customers.

Company A, the largest of the participating companies, also drew attention to its scientific knowledge and shared its ability to combine this with its technical, commercial and managerial expertise in building value-creating relationships (Table 7, quote 30). It was through a process of sustained, medium to long-term relationship building that Company A was able to share its scientific knowledge to build bonding social capital through interactions with individuals and groups with a close connection. Social capital accumulated through these established ties creates value and adds to commercial success (Garud, Gehman and Giuliani, 2018). When dealing with new potential customers, Company A would invite them to visit the plant to demonstrate its knowledge and understanding of its science and technology to build credibility (Table 7, quote 31). In another case, Company C offered free services such as testing, validation and reporting services, in the hope that the company's scientific and technical knowledge and expertise would act as a bridge, strengthening chances

to 'get ahead' through building relationships that create value by generating new contracts (Table 7, quote 32).

For Companies A and C, forging relationships with existing and potential customers by showcasing their scientific knowledge and expertise, and offering free testing and validation services, was essential to building social capital. Other companies combined approaches, using existing connections to identify potential new customers. The Managing Director of Company B, for example, combined bonding and bridging approaches by 'carrying over' his existing good relations with customers to employees, thus acting as a bridge that was critical to the success of the company (Table 7, quote 33). Similarly, Company F benefited from strong bonding social capital through good relations with existing customers who passed on recommendations, a form of bridging social capital, providing avenues to new potential customers (Table 7, quote 34). Our fieldwork demonstrates the significance of relationship-building processes and the place of institutional intermediaries in servicing the development and exploitation of social capital in the commercialization of new ideas and the establishment of new collaborative projects. Interactions associated with bonding and bridging activities are central to the development of social capital and a core capability of effective boundary spanners.

Discussion: building social capital and value creation in science-based SMEs

Our research offers insight into boundary-spanning activities and the processes of sensemaking that create knowledge and build meaningful relationships (Hettich and Kreutzer, 2021). The findings highlight how boundary spanners assimilate knowledge from a range of sources and, through interactions with others, make sense in seeking to integrate and recombine knowledge (Newell, Tansley and Huang, 2008). These processes generate social capital that serves as a knowledge resource for identifying and evaluating ideas and opportunities. Some of these ideas are taken up in the form of new collaborative projects that create value for science-based SMEs. Central to these processes is the ability to establish and sustain meaningful relationships, whilst also drawing upon the knowledge and expertise of close colleagues and friends who form part of historical and long-established networks. We highlight how boundary spanners act as collective agents, relating practices across disciplines through established and negotiated relationships (Banerjee and Corredoira, 2013) that support knowledge exchange and recombination (Wang, Chen and Fang, 2018). In our analysis of boundary-spanning activities that support the building and maintenance of social capital, there are some interesting dif-

ferences and similarities at the individual and collective level.

Our findings reveal the importance of collective sensemaking (Maitlis and Christianson, 2014) and, significantly, demonstrate how temporal sensemaking facilitates collaboration across organizational and disciplinary boundaries. Collaboration enables some boundary spanners to move between the backward glance (retrospective) (Weick, Sutcliffe and Obstfeld, 2005) and more future-oriented (prospective) sensemaking (Konlechner *et al.*, 2019). We illustrate how the understanding and use of scientific knowledge is influenced by the sensemaking of individuals and groups that is largely retrospective (backward glancing) or prospective (future oriented). Figure 3 provides a representational framework delineating some of these differences.

Our framework identifies four ways that organizations use scientific knowledge to build social capital. Unsurprisingly, bonding social capital (Quadrant 1) was used by individual boundary spanners who employed sensemaking processes triggered through historical connections and personal contacts with ex-colleagues (Pattinson, Nicholson and Lindgreen, 2018). At an individual level, bridging social capital (Quadrant 2) supported a more future-oriented sensemaking in which boundary spanners attempted to use scientific knowledge to build rapport with potential customers, a key component in value creation through knowledge integration and combination (Wang, Chen and Fang, 2018). Linking social capital (Quadrant 3) was mostly associated with collective boundary spanning that used scientific knowledge from established institutional intermediaries and professional connections. Our findings suggest that some boundary spanners attempted to create new ties to powerful institutional actors, such as universities and government connections, which we call *mediating* social capital (Quadrant 4). Linking plays a central role, especially among our smaller start-up SMEs (Companies C and D), who lacked the boundary-spanning capabilities of more established SMEs. For the more established SMEs (e.g. Company E), mediating social capital enabled them to create ties to more powerful partners, such as the US government.

In practice, organizations exhibit various overlapping configurations arising from activities within more than one quadrant. These are open to continual modification and change. However, there is likely to be a shift in the focus of organizations as circumstances alter and new opportunities present themselves. An effective configuration comprises all elements in being able to combine bonding, bridging, linking and mediating activities, seeking to secure value creation (see Table 8 for a cross-case synthesis of these findings). Building social capital through individual and collective processes of temporal sensemaking is central in building knowledge capabilities (Kogut and Zander, 1992). As we show,

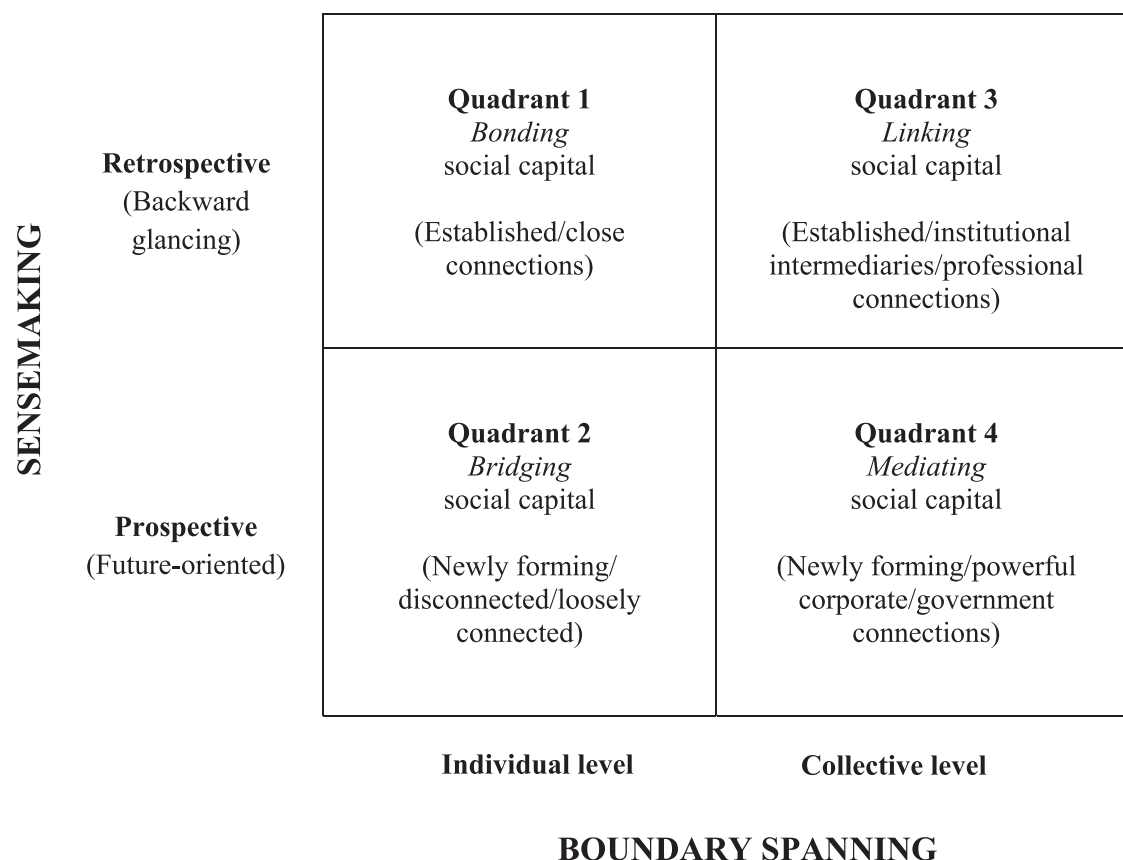


Figure 3. Representational framework delineating processes of sensemaking and boundary-spanning activities in the building of social capital

these capabilities rely on boundary-spanning activities across disparate scientific and technological disciplines as well as mediating relationships with powerful institutions (Rosenkopf and Nerkar, 2001). Our findings provide significant insight into these processes, which can flourish among diverse scientific and related disciplines, enabling the exchange and recombination of knowledge that supports value creation (Banerjee and Corredoira, 2013).

In clarifying these relationships, Figure 4 presents an inductive model charting the dynamic processes that generate value creation in science-based SMEs.

Our model outlines the dynamic interlacing of boundary-spanning activities, sensemaking processes and the bonding, bridging, linking and mediating elements of social capital that supports value creation. For example, longstanding employees, such as founders, were shown to draw extensively on past experience and historical connections. Although only a small sample, the inductive generalization that founders use established networks and historical friendships/connections in seeking to make sense of new scientific knowledge that supports ongoing learning and knowledge acquisition is reasonable.

There are also causal inferences, for instance, prospective sensemaking is prevalent given the premise that

science-based SMEs constantly search for new insights and knowledge to support the commercialization of innovation. In other words, inductive reasoning that future-oriented thinking – in seeking to synthesize new knowledge to shape future outcomes – is an essential component of science-based SMEs. Additionally, from illustrating how boundary spanners work to make sense of others when establishing and building relationships, some boundary spanners used temporal sensemaking to move between the backward-glance (retrospective) and more future-oriented (prospective) sensemaking. In the example of science-based SMEs, we can further analytically deduce that such sensemaking will necessarily be temporal, including prospective as well as retrospective sensemaking, given the future orientation of such companies.

The processual dynamics among the main components we identify for value creation in science-based SMEs are highlighted in our empirical analysis. Individual and collective boundary spanning is integral to building and sustaining relationships among individuals who are closely connected (bonding) and disconnected, or loosely connected (bridging) by disciplinary, cultural and technological backgrounds. It also enables opportunities for promoting the development of social capital through linking to institutions and other external

Table 8. Cross-case synthesis of findings

Finding	Company A	Company B	Company C	Company D	Company E	Company F	Company G
Quadrant 1 – Bonding social capital (established/close connections)							
Sharing scientific expertise/knowledge with established customers	X	X	X				
Leveraging personal connections				X	X		X
Reliance on historic connections/old friends				X	X		X
Accessing ex-colleagues				X			
Knowledge recombination across scientific and technological boundaries	X				X		
Quadrant 2 – Bridging social capital (newly forming/disconnected/loosely connected)							
Building bridging social capital between sales staff and customers					X		
Free testing and validation services to ‘get ahead’			X				
Using peripheral knowledge to build ties		X					
Creating mutual benefit across organizational domains						X	
Customer recommendations as a bridge to potential new customers						X	
Organizational reputation as a bridge to potential new customers							X
Quadrant 3 – Linking social capital (established/institutional intermediaries/professional connections)							
Connecting with [N] Central as a ‘dating agency’, creating links to other nanotechnology companies				X			
Using the [W] Innovation Centre’s forum as an institutional intermediary			X				
Quadrant 4 – Mediating social capital (newly forming/powerful corporate/governmental connections)							
Mediating relationships that create ties to more powerful partners, such as the US government					X		
Overlapping configurations – combining bonding, bridging, linking and mediating social capital (established/newly forming/close and loose connections)							
Carrying over good relations that combine bonding, bridging, linking and mediating approaches							
Building on established networks and forming new alliances			X			X	

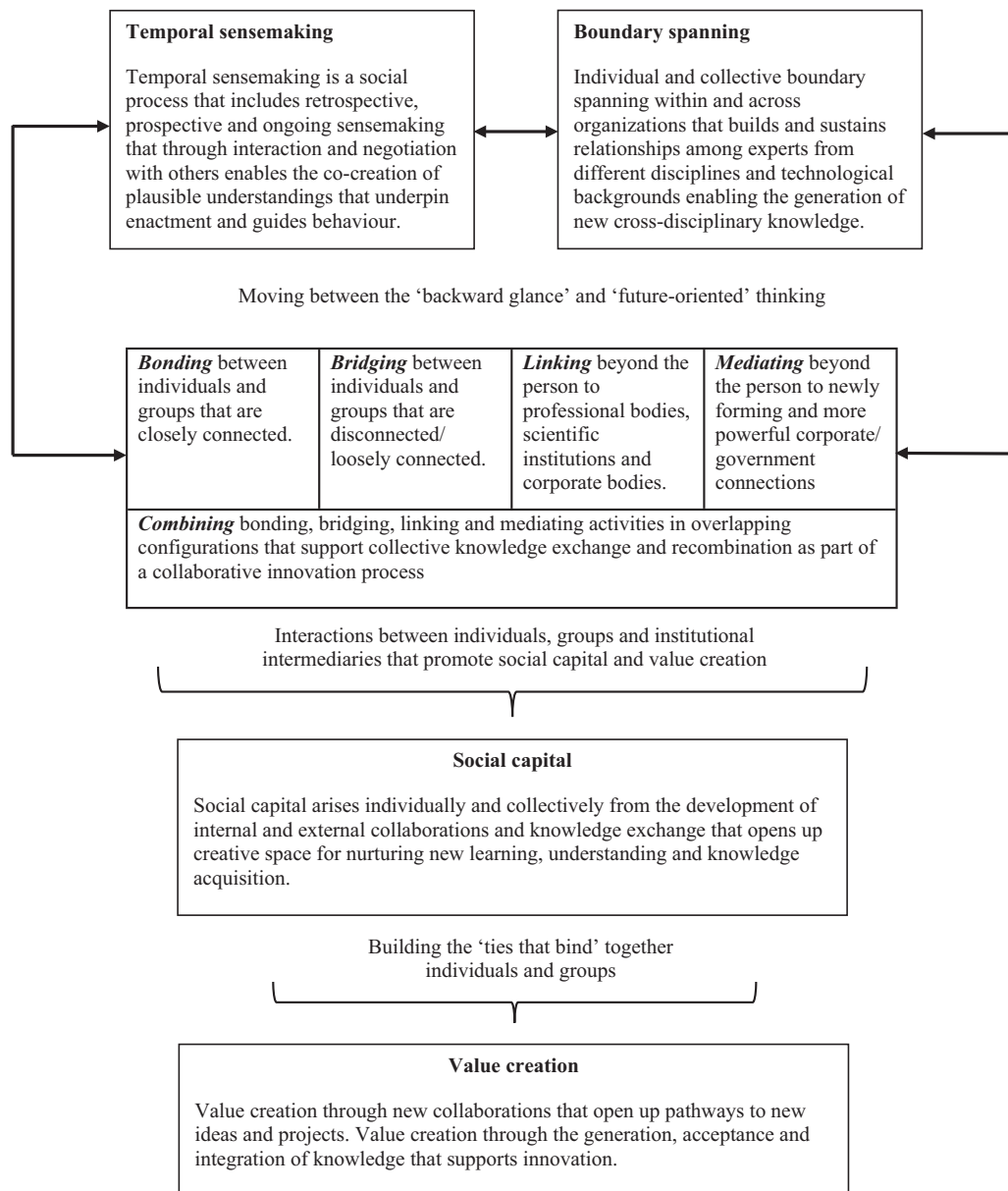


Figure 4. Inductive model of value creation in science-based SMEs

bodies, or by creating ties to more powerful institutions that exploit mediating social capital. During these dynamic social processes, individuals continuously strive to make sense of what is going on around them. As our data underlines, this involves sensemaking in the here and now that is ongoing, drawing on both the past (retrospective) and future (prospective). New understandings arise that, if plausible, offer avenues for action guiding behaviour in further collaborative engagement that identify, test, develop and assess future innovative projects.

Conclusion and contributions

There are a number of theoretical and conceptual insights that arise from our research. First, the importance of boundary spanners and the need to extend our research gaze to include collective boundary-spanning activities within and across organizations. In exploring these dynamics, our research extends our understanding of how boundary spanners integrate and recombine scientific expertise through sensemaking processes that influence collective action in solving customer

problems and generating new ideas for project development (Banerjee and Corredoira, 2013; Marshall and Rollinson, 2004). We demonstrate how they hold a key role in bringing together and utilizing knowledge that crosses boundaries and influences others in promoting the plausibility of new projects and collaborations. They exploit serendipitous opportunities and peripheral knowledge to create ties to existing and new potential customers. They act as a reservoir of boundary-spanning expertise, passing on their success and knowledge to less experienced colleagues. This was particularly critical in SMEs that lack individual boundary-spanning ability. In short, they are central protagonists in the creation of value in science-based SMEs.

Second, the centrality of sensemaking in co-creating plausible narratives that people can engage with in building relationships and developing future collaborations. We demonstrate the need to go beyond a conceptual focus on retrospective sensemaking to include not only prospective sensemaking but also the interplay of past, present and future in accommodating processes of temporal sensemaking. Retrospective sensemaking centres on the knowledge, skills and experiences of engaging over a prolonged period of time and was often triggered through historical connections and personal contacts. Prospective sensemaking processes were more in evidence during the search for new customers and in evaluating future potential markets. We show, for example, how prospective sensemaking enabled boundary spanners to decipher complex scientific and technological knowledge that helped shape future outcomes. Temporal sensemaking further extends this concept, drawing attention to how people interpret the past and expectations of the future in making sense of an ongoing present, and how this continual interplay of past, present and future shapes narratives that inform action (Dawson and Sykes, 2018). These processes of temporal sensemaking help boundary spanners build social capital through developing and expanding a network of relations that combines long-established relationships and past experience with new customers, collaborations and institutional connections.

Third, the need for bonding, bridging, linking and mediating activities in building social capital. In our examination, we identify the need to go beyond an individual focus in accommodating groups and collectivities, and to distinguish between linking activities (that extend beyond the person to professional bodies, scientific institutions and corporate bodies) and what we term mediating activities (those that extend beyond the person to newly forming and more powerful corporate/government connections). We also highlight how these often overlap and combine in configurations that support collective knowledge exchange and recombination as part of a collaborative innovative process.

Our inductive model (Figure 4) incorporates these additional elements in attending to the conventional concern with individual boundary-spanning activities and retrospective sensemaking, whilst also including collective boundary-spanning activities and prospective sensemaking. Our research shows how contexts that combine individual and collective boundary spanning through a continual interplay of retrospective and prospective sensemaking are most useful for exploiting scientific knowledge that builds social capital and creates value. We show how, through sustained, medium to long-term relationship building, companies were able to exploit bonding and bridging social capital among existing networks, opening up opportunities to identify new customers and forge new collaborative relationships. Our findings also reveal how institutional intermediaries can act as a 'dating agency', linking social capital in making introductions that help companies make new connections. In addition, our findings show how mediating social capital was deployed in actively seeking more powerful institutional partners. This network of relations supports the potential exploitation of cross-disciplinary scientific knowledge through temporal sensemaking processes. These sensemaking processes support the co-creation of new ideas and understandings that present a platform for action in guiding behaviour. In this, our findings ably demonstrate how boundary spanners engage in sensemaking in their interactions across and within organizations. Boundary spanners used bonding, bridging, linking and mediating activities that promote the development of social capital and contribute to value creation in science-based SMEs.

Managerial implications

The evidence arising from this research raises implications for managers in science-based SMEs. First, our findings enable managers to better understand how individuals make sense of the world around them. They highlight how individuals use their experiences to enact boundary-spanning activities that create bonding, bridging, linking and mediating activities which generate social capital, central to the process of value creation. Second, managers should recognize that effective boundary spanners often leverage future value from historic connections. Where they cannot access historic connections, managers might consider seeking boundary-spanning support to mediate relationships. This might be by creating ties to more powerful institutional or industry-specific intermediaries to help them exploit scientific knowledge in their search for new innovations and collaborative opportunities. Third, managers should recognize that as individuals achieve managerial seniority and take on new roles and responsibilities, they may find it more challenging to participate in boundary-spanning

activities. To counter this problem, managers need to consider providing opportunities for senior managers to network with external people, for example by attending industry events and/or conferences. This will help mitigate the impact of seniority on boundary-spanning effectiveness.

Future research

Our findings open up some interesting avenues for further research, especially in relation to collective boundary spanning. Whilst we did find examples of collective action, the mechanisms for participation in collective action would benefit from further exploration. Our inductive model provides a useful point of departure. However, further theoretical development is also required to explore the effectiveness of different configurations that emerge as boundary spanners seek to build particular types of social capital to support their strategic objectives. We also contend that the importance of temporal sensemaking to boundary-spanning activities that create organizational value remains an area worthy of further study.

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